



PCF lab Certificate Rotation Estimate

- **Director:** 192.168.2.2
- **Generated:** 2026-06-10 10:13:00 UTC
- **Horizon:** certificates expiring within 5y
- **Topology source:** maestro tp
- **Update rules:** bosh manifest (per-IG max_in_flight/canaries)

Summary

Metric	Value
Expiring certs ($\leq 5y$)	112 (✗ 0 · ⚠ 0)
— leaf / CA	105 / 7
Live VMs (DB nodes)	7 (2)
One foundation-wide Apply	1h 00m – 1h 50m
Estimated rotation time	11h 00m – 20h 10m

Foundation inventory

Deployment	Type	VMs	DB nodes
cf-dbe1a7580979a87638e7	cf	7	2

2 instance group(s) have `serial:false` — they may update in parallel, so real time can come in under the (conservative serial) estimate.

CA rotations (3-phase)

CA	Scope model	Apply plan	Expires	Severity
<code>.properties.nats_client_ca.3506e33356bef86af963</code>	FOUNDATION	3x foundation-wide	2030-05-29T03:06:57Z	 info
<code>.properties.root_ca.3506e33356bef86af963</code>	FOUNDATION	3x foundation-wide	2030-05-29T03:06:57Z	 info
<code>/p-bosh/cf-dbe1a7580979a87638e7/diego-instance-identity-intermediate-ca-2-7</code>	DEPLOYMENT	3x cf-dbe1a7580979a87638e7	2028-06-09T04:19:47Z	 info
<code>/opsmgr/bosh_dns/tls_ca</code>	FOUNDATION	3x foundation-wide	2030-05-29T16:32:50Z	 info
<code>/cf/diego-instance-identity-root-ca-2-6</code>	DEPLOYMENT	3x cf-dbe1a7580979a87638e7	2029-05-29T16:32:45Z	 info
<code>/services/tls_ca</code>	TRUSTED	2x foundation-wide (BOSH trusted-certs add+remove) + 1x cf-dbe1a7580979a87638e7	2031-05-29T03:24:34Z	 info
<code>opsman-root-ca:3506e333</code>	FOUNDATION	3x foundation-wide	2030-05-29T03:06:57Z	 info

Leaf certificates (1-phase)

Deployment	Leaf certs	Apply
<code>p-bosh (not VM-counted)</code>	11	1× (with the leaf campaign)
<code>cf-dbe1a7580979a87638e7</code>	94	1× (with the leaf campaign)

Estimate breakdown

Campaign	Applies	Time (low – high)
Leaf certs	1× over 2 deployment(s)	1h 00m – 1h 50m
CA certs (batched)	3× foundation-wide + per-deployment leaf-regen	10h 00m – 18h 20m
Total		11h 00m – 20h 10m

Model & assumptions

- 20m overhead per Apply Changes; stateless 4–10m/VM, DB 10–20m/node.
- A FOUNDATION CA = 3 foundation-wide applies; a services TLS CA = 2 foundation + 1 deployment applies; a deployment CA = 3 on its deployment.
- Estimate is **Apply Changes compute time only** — it excludes change-window / approval gaps between phases, which often dominate the wall-clock for CA rotations.